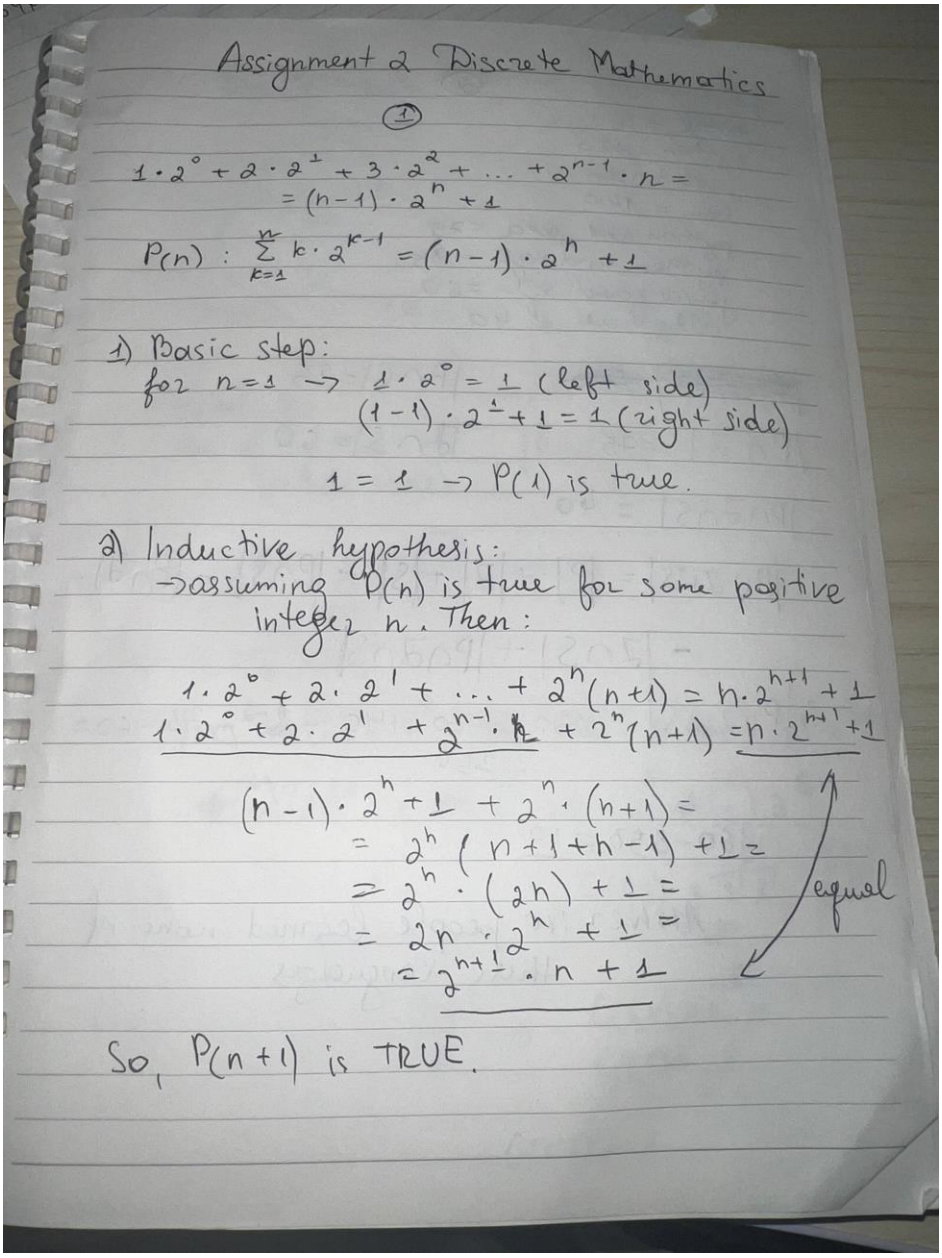
YIGITALIEVA MUBINA 20253721 DISCRETE MATHEMATICS ASSIGNMENT 2



Assignment 2 Discrete Mathematics

①

$1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \dots + 2^{n-1} \cdot n = (n-1) \cdot 2^n + 1$

$P(n): \sum_{k=1}^{n} k \cdot 2^{k-1} = (n-1) \cdot 2^n + 1$

1) Basic step:
for $n=1 \rightarrow 1 \cdot 2^0 = 1$ (left side)
$(1-1) \cdot 2^1 + 1 = 1$ (right side)

$1 = 1 \rightarrow P(1)$ is true.

2) Inductive hypothesis:
→assuming $P(n)$ is true for some positive integer n. Then:

$1 \cdot 2^0 + 2 \cdot 2^1 + \dots + 2^n(n+1) = n \cdot 2^{n+1} + 1$

$1 \cdot 2^0 + 2 \cdot 2^1 + 2^{n-1} \cdot n + 2^n(n+1) = n \cdot 2^{n+1} + 1$

$(n-1) \cdot 2^n + 1 + 2^n \cdot (n+1) =$
$= 2^n (n+1+n-1) + 1 =$
$= 2^n \cdot (2n) + 1 =$
$= 2n \cdot 2^n + 1 =$
$= 2^{n+1} \cdot n + 1$

equal

So, $P(n+1)$ is TRUE.

②

total = 360
python = 220
java = 170
SQL = 140
python and java = 75
python and SQL = 85
java and SQL = 60
all three = 40

~~|P ∩ J ∩ S|~~ $|P \cap S| = 85$

$|P \cap J| = 75$ $|J \cap S| = 60$

$|P \cap J \cap S| = 40$

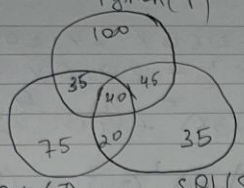
$$|P \cup J \cup S| = |P| + |J| + |S| - |P \cap S| - |P \cap J| - |J \cap S| + |P \cap J \cap S|$$

$$|P \cup J \cup S| = 220 + 170 + 140 - 85 - 75 - 60 = 350$$

$360 - 350 = \underline{10}$

Answer: 10 people learned none of these languages.

Venn Diagram



Given:

$|P| = 220$
$|S| = 140$
$|J| = 170$

$|P \cap J| = 75$
$|P \cap S| = 85$
$|J \cap S| = 60$
$|P \cap J \cap S| = 40$

Total = 360
none of the three = 10
(outside circle).

a) $(2x-3)^6 = \sum_{j=0}^{6} \binom{6}{j} (2x)^{6-j} 3^j$ (3)

$C(n,2) = \frac{n!}{2!(n-2)!}$

$\binom{6}{0}(2x)^6 - \binom{6}{1}(2x)^5 \cdot 3 + \binom{6}{2}(2x)^4 \cdot 3^2 - \binom{6}{3}(2x)^3 3^3 +$

$+ \binom{6}{4}(2x)^2 \cdot 3^4 - \binom{6}{5} 2x \cdot 3^5 + \binom{6}{6} 3^6$

$C(6,0) = \frac{6!}{0!\,6!} = 1$

$C(6,1) = \frac{6!}{1!\,5!} = 6$

$C(6,2) = \frac{6!}{2!\,4!} = \frac{5 \cdot 6^3}{2} = 15$

$C(6,3) = \frac{6!}{3!\,3!} = \frac{4 \cdot 5 \cdot 6}{2 \cdot 3} = 20$

$C(6,4) = \frac{6!}{4!\,2!} = 15$

$C(6,5) = \frac{6!}{5!\,1!} = 6$

$C(6,6) = 1.$ →

$= 64x^6 - 32 \cdot 6x^5 \cdot 3 + 15 \cdot 16x^4 \cdot 9 - 20 \cdot 8x^3 \cdot 27 + 15 \cdot 4x^2 \cdot 81 - 12x \cdot 81 \cdot 3 + 729 =$

$= 64x^6 - 576x^5 + 2160x^4 - 4320x^3 + 4860x^2 - 2916x + 729.$

b) $(3x+2y)^5 = \sum_{i=0}^{5} = \binom{5}{0}(3x)^5 + \binom{5}{1}(3x)^4 \cdot 2y +$

$+\binom{5}{2}(3x)^3 \cdot (2y)^2 + \binom{5}{3}(3x)^2 (2y)^3 + \binom{5}{4} 3x \cdot (2y)^4$

$+ \binom{5}{5}(2y)^5$

$C(5,0) = \frac{1}{1}$

$C(5,1) = \frac{5!}{1! \, 4!} = 5$

$C(5,2) = \frac{5!}{2! \, 3!} = \frac{4 \cdot 5}{2} = 10$

$C(5,3) = \frac{5!}{3! \, 2!} = 10$

$C(5,4) = \frac{5!}{4! \, 1!} = 5$

$C(5,5) = 1.$

$(3x)^5 + 5 \cdot (3x)^4 \cdot 2y + 10 \cdot (3x)^3 \cdot (2y)^2 + 10 \cdot (3x)^2 \cdot (2y)^3 + 5 \cdot 3x \cdot (2y)^4 + (2y)^5 =$

$= 243x^5 + 810x^4 y + 1080x^3 y^2 + 720x^2 y^3 + 240xy^4 + 32y^5$